Software Development for AI

EEE 5653

Final Presentation

Damage Classification and Localization for Composite Plates with Temperature Variance Using Deep Learning

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Introduction

- Carbon fiber-reinforced polymers are used across many different fields.
- These materials are usually under a variety of conditions; temperature would be the most common one to change.
- Non-destructive testing is a common approach to detect defects for carbon fiber-reinforced polymers.

Theoretical Framework

- Baseline Subtraction

- Baseline Selection is a method that subtracts a reference signal from the analyzed signal or vice versa
- This should remove similarities in the two objects, like structural elements
- The issue of this method is that with differing environments, the signals are different due to material properties of the two pieces

Theoretical Framework

- Optimal Baseline Selection

- Optimal Baseline Selection (OBS) is selecting a baseline signal that has similar Temperatures to the analysed signal
- A collection of reference signals are collected to then compare based on there temperature
- Below is a Bad example where the temperature of the 2 samples is significantly different

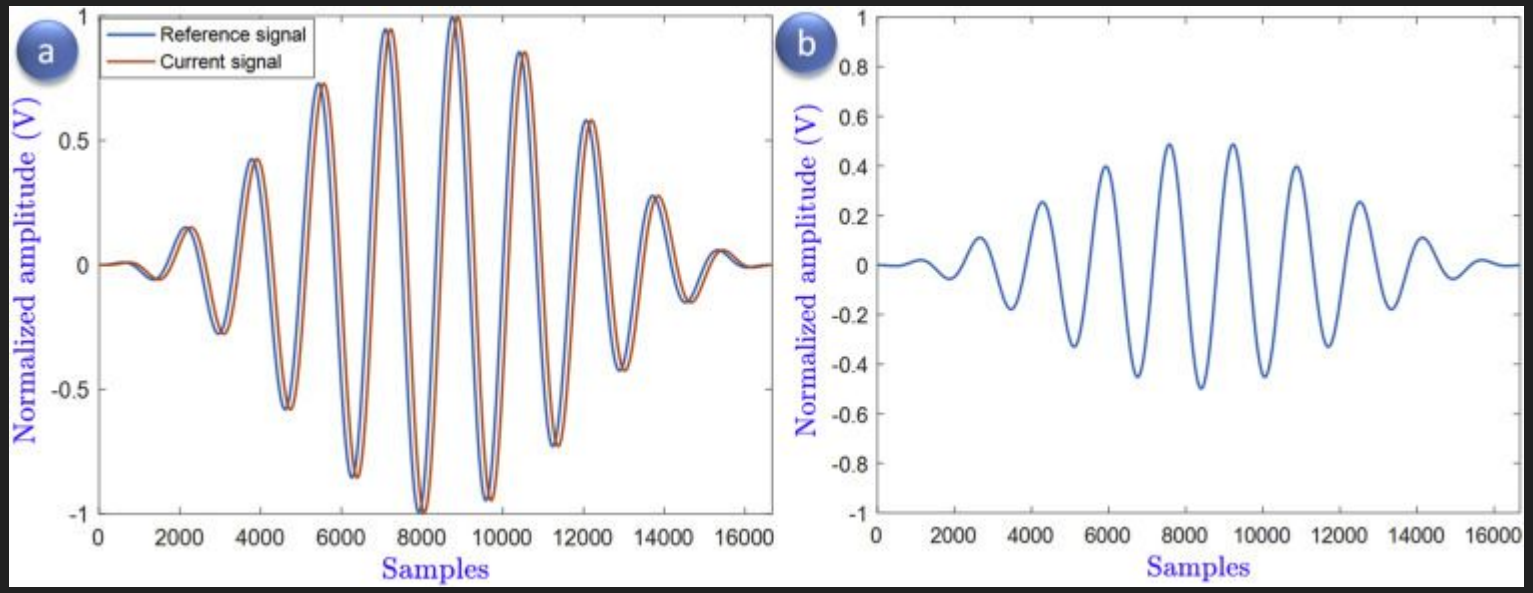


Figure 1: Bad Baseline from [9]

Theoretical Framework

- Baseline Signal Stretch (BSS)

- BSS is used to stretch a Baseline signal small amounts
- Used in structural health monitoring for EOC effects
- Takes a baseline signal and stretches it by stretch factor Alpha
- 2 ways I have tried are optimization function to find the minimum phase difference between stretched baseline and current signal
- another uses cross-correlation.

Theoretical Framework

- Wave Speed Cross Correlation

- Wave speed can be found by taking the lag between the cross-correlation of the input and output signal
- Cross-correlation can be found similarly to a convolution shown below for 2 one-dimensional arrays below
- It is found using Once a cross-correlation is found, the lag between the two can be found
- After the distance between the sensors and the lag can be used to find the wave speed

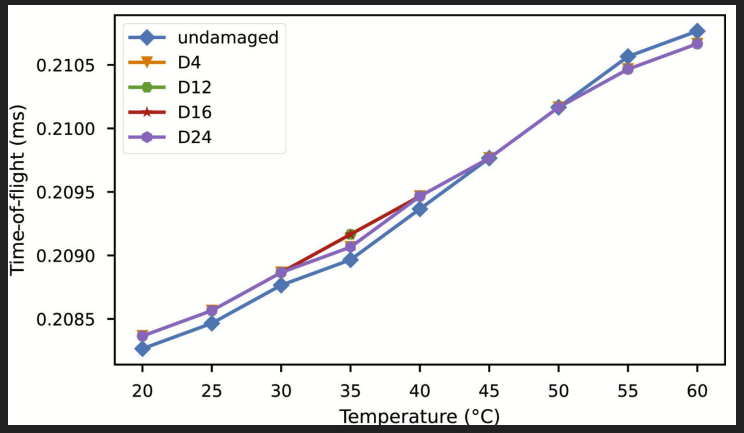


Figure 2: Relationship of Temperature and class from [12]

$$z[k] = \sum_{l=0}^{N-1} x_l \, y^*_{l-k}$$

Figure 3: Equation of Cross Correlation from [11]

Theoretical Framework

- Periodogram and Kaiser Window

- Periodograms are computed using the Discrete Fourier Transform as shown below
- They give the Power spectral Density of a Time series
- This is the Magnitude of a DFT squared

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

Figure 4: Equation of Discrete Fourier Transform

Theoretical Framework

- Convolutional Neural Network (CNN)

- Used to add locational context to Deep Learning features
- Used primarily in images
- Layers of microfeatures into macrofeatures

A	B	C	D	E	F	G	H	I	J
Z	Y	X	W	V	U	T	S	R	Q

Table 1: Example CNN array

Theoretical Framework

- Multi-Layer Perceptron (MLP)

- Multilayer Perceptrons are forward-propagating neural networks
- Function on a weight sum that is fed into activation function
- Loss is calculated per weight and bias through backpropagation
- Weights and bias are modified to reduce loss in each epoch.

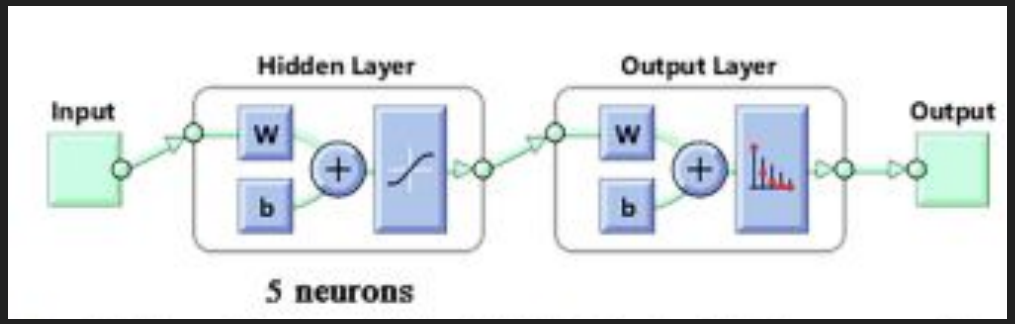


Figure 5: Multilayer Perceptron example adapted from [2]

Theoretical Framework

- Dimentionality Reducers

- PCA is a Dimensional Reduction technique
- Principal Component 1 is the direction in feature space with the most variance
- Principal Component 2 is the direction in feature space with the second most variance, but must be unrelated to the first
- Any further components must be similar to 2 in that all other components must be unrelated to them.

Theoretical Framework

- Clustering

- Clustering techniques, more specifically DB-Scan
- DBScan takes points within a maximum range, eps, and requires a certain amount of points; otherwise, they are classified as noise

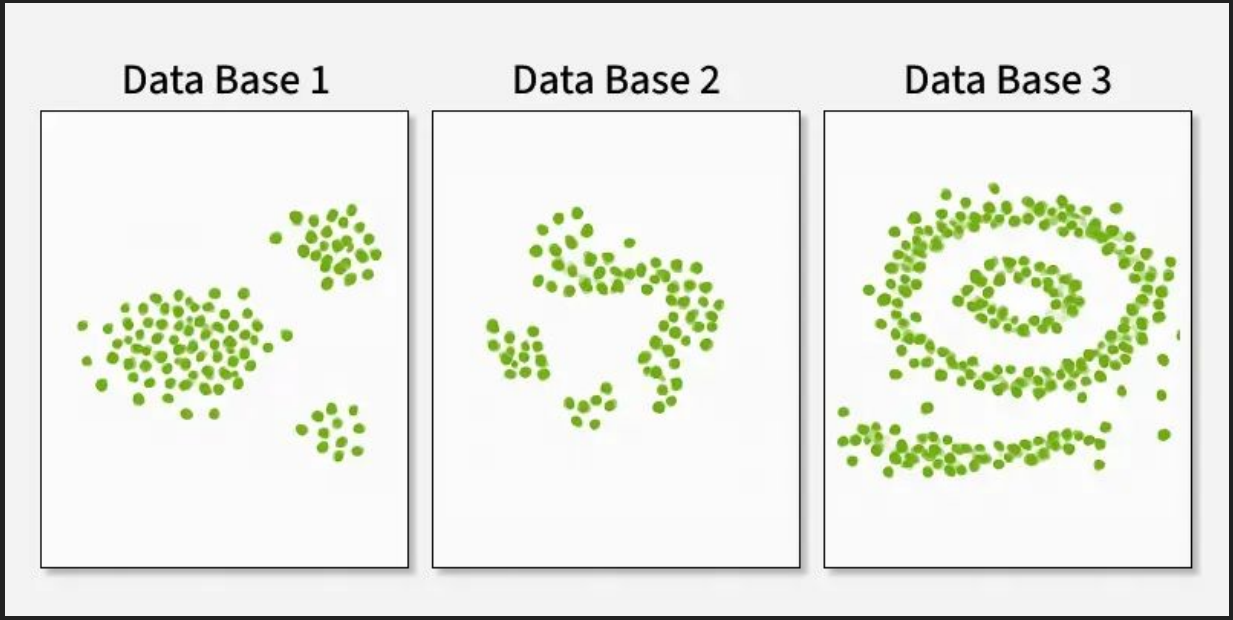


Figure 6: Multilayer Perceptron example adapted from [10]

Problem Statement

- Materials are commonly damaged in forms that are unviewable
- Guided wave signal processing and extracted features can be accurately used for detection and classification of damage

Experimentation

- Experiments were performed on the Open Guided Waves Dataset 2, which has temperature variance and 966 total measured plates
- The dataset has 322 baseline and 161 of each of 4 damage locations
- Of these, the 80 in the "training set" are used for OBS

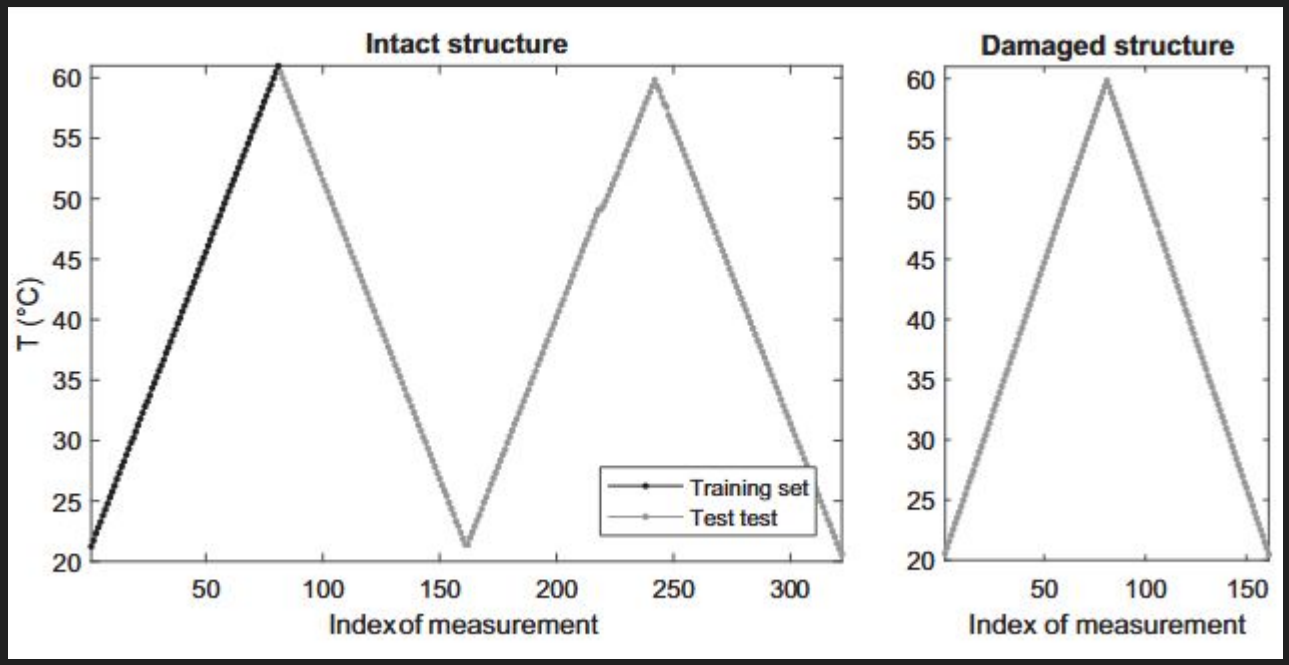


Figure 7: Temperatures of Samples adapted from [3]

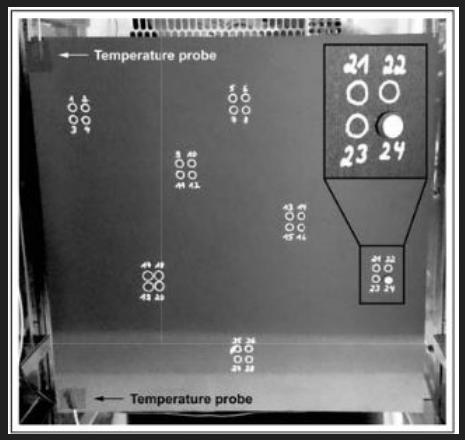


Figure 8: Experimental setup with damage locations adapted from [3]

Methodology / Project Design

- Principal Component Analysis (PCA)

- I have tried multiple different methods with PCA
- I tried 2 different ways of classifying
  - One was fed into an MLP; after 10 epochs, it was able to get 70 percent accuracy. This was fed everything.
  - The other was a custom feature map involving the wave speed, wave magnitudes, and frequency magnitudes.

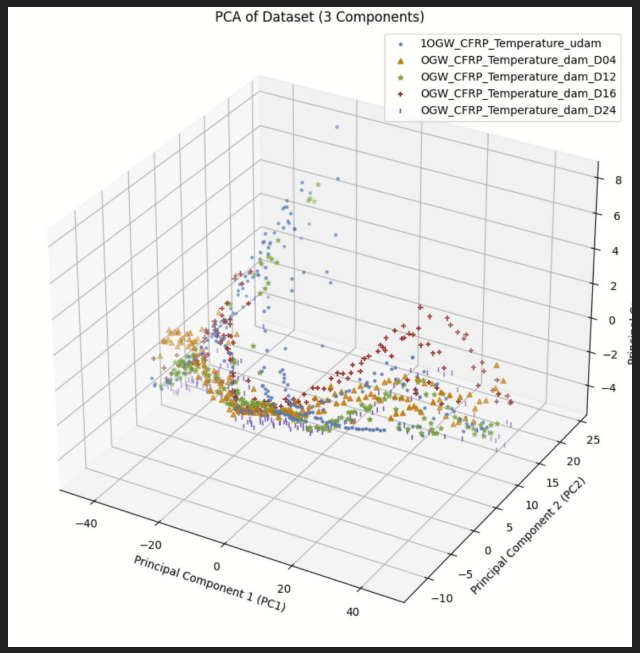


Figure 9: PCA with Class labels

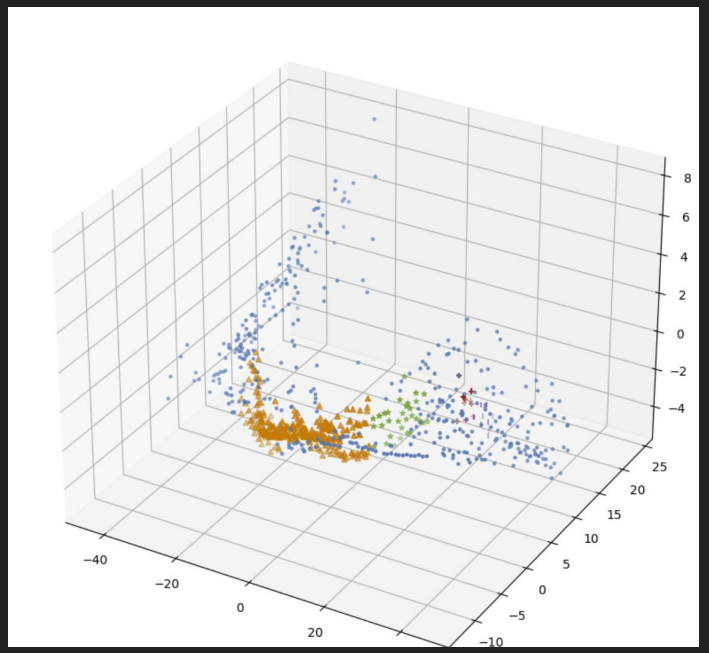


Figure 10: DBSCANS labeling in PCA dimensions

Methodology / Project Design

- CNN to MLP

  - There were 3 models made starting from super stack. I didn't realize how many parameters were made, so I downsized from there.
  - The initial model was made to try and get the wave speed, possibly with enough Conv2Ds, but I removed it to attempt to get rid of errors.
  - The CNN
    - 2 layers of Conv2Ds 2x10
    - Into max pooling 2x2, the last Max pooling is a 1x2
  - Then MLP to classify
    - initially 5 layers of 512 neurons halving until the classifier layer (super)
    - reduced to 2 layers and changed the neurons between the other 2 (normal and micro)
  - All models go through 50 Epochs

Methodology / Project Design

- Super Stack

Layer (type)	Output Shape	Param #
conv2d_8 (Conv2D)	(None, 5, 13099, 32)	672
max_pooling2d_8 (MaxPooling2D)	(None, 2, 6549, 32)	0
conv2d_9 (Conv2D)	(None, 1, 6540, 64)	41,024
max_pooling2d_9 (MaxPooling2D)	(None, 1, 3270, 64)	0
flatten_4 (Flatten)	(None, 209280)	0
dense_16 (Dense)	(None, 512)	107,151,872
dense_17 (Dense)	(None, 256)	131,328
dense_18 (Dense)	(None, 128)	32,896
dense_19 (Dense)	(None, 64)	8,256
dense_20 (Dense)	(None, 5)	325

Table 2: Super Stack Model Summary

Figure 11: Super Stack Trainable Params

```
Total params: 322,099,121 (1.20 GB)

Trainable params: 107,366,373 (409.57 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 214,732,748 (819.14 MB)
```

Results and Discussion

Super Stack

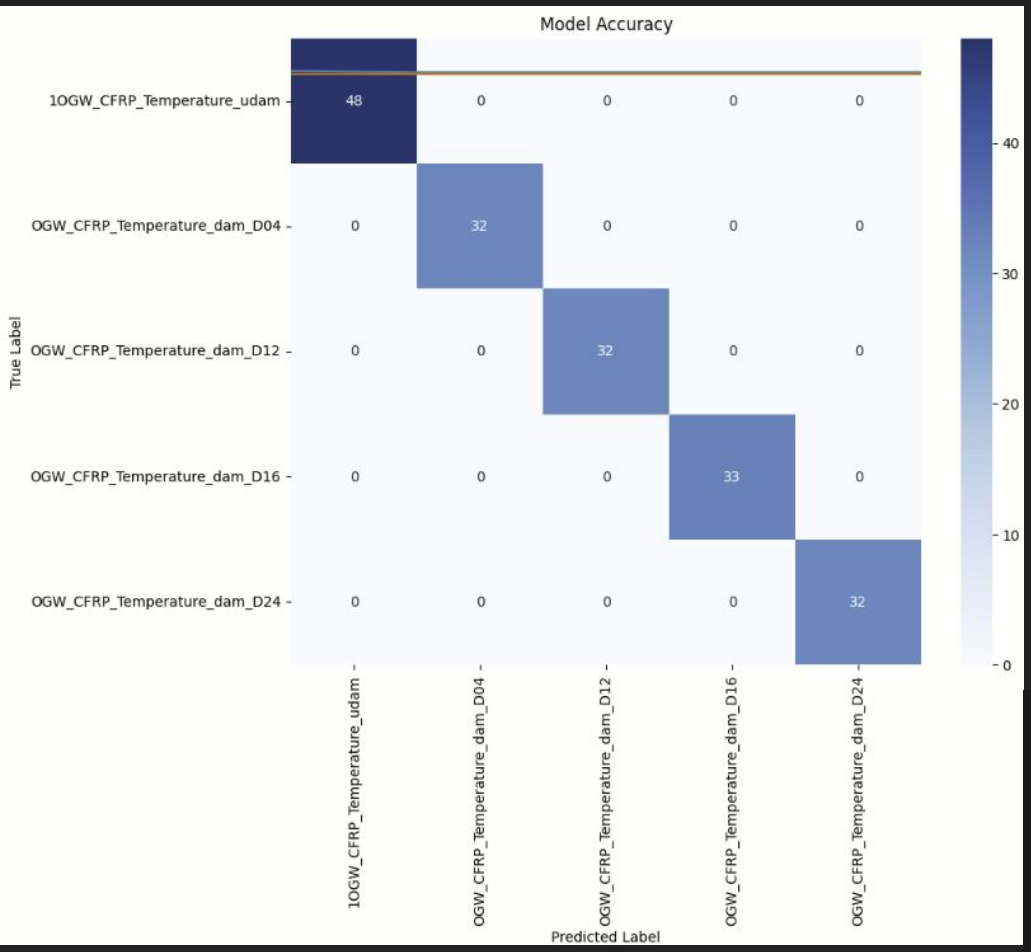


Figure 12: Super Stack Model Confusion Matrix

Time to Train CPU
Only: ~15 minutes

Table 3: Super Stack Statistics

	precision	recall	f1-score	support
0	1.00	1.00	1.00	48
1	1.00	1.00	1.00	32
2	1.00	1.00	1.00	32
3	1.00	1.00	1.00	33
4	1.00	1.00	1.00	32
accuracy			1.00	177
macro avg	1.00	1.00	1.00	177
weighted avg	1.00	1.00	1.00	177

Methodology / Project Design

- Stack

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 5, 13099, 32)	672
max_pooling2d_4 (MaxPooling2D)	(None, 2, 6549, 32)	0
conv2d_5 (Conv2D)	(None, 1, 6540, 64)	41,024
max_pooling2d_5 (MaxPooling2D)	(None, 1, 3270, 64)	0
flatten_2 (Flatten)	(None, 209280)	0
dense_10 (Dense)	(None, 128)	26,787,968
dense_11 (Dense)	(None, 64)	8,256
dense_12 (Dense)	(None, 5)	325

Table 4: Stack Model Summary

Figure 13: Stack Trainable Params

Total params: 80,514,737 (307.14 MB)

Trainable params: 26,838,245 (102.38 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 53,676,492 (204.76 MB)

Results and Discussion

Stack

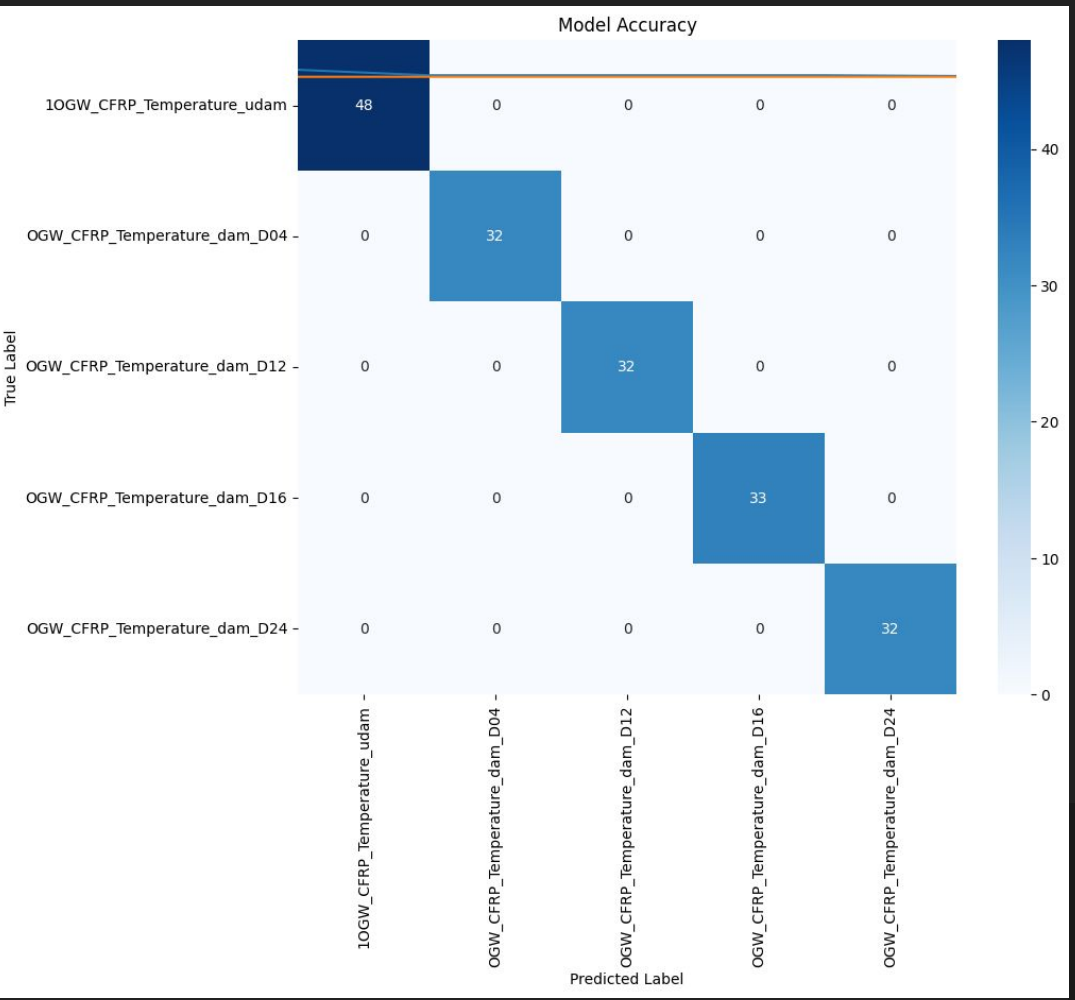


Figure 14: Stack Model Confusion Matrix

Time to Train CPU
Only: ~10 minutes

Table 5: Stack Statistics

	precision	recall	f1-score	support
0	1.00	1.00	1.00	48
1	1.00	1.00	1.00	32
2	1.00	1.00	1.00	32
3	1.00	1.00	1.00	33
4	1.00	1.00	1.00	32
accuracy			1.00	177
macro avg	1.00	1.00	1.00	177
weighted avg	1.00	1.00	1.00	177

Methodology / Project Design

- Micro Stack

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 5, 13099, 32)	672
max_pooling2d_6 (MaxPooling2D)	(None, 2, 6549, 32)	0
conv2d_7 (Conv2D)	(None, 1, 6540, 64)	41,024
max_pooling2d_7 (MaxPooling2D)	(None, 1, 3270, 64)	0
flatten_3 (Flatten)	(None, 209280)	0
dense_13 (Dense)	(None, 64)	13,393,984
dense_14 (Dense)	(None, 32)	2,080
dense_15 (Dense)	(None, 5)	165

Table 6: Stack Model Summary

Figure 15: Super Stack Trainable Params

```
Total params: 40,313,777 (153.78 MB)

Trainable params: 13,437,925 (51.26 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 26,875,852 (102.52 MB)
```

Results and Discussion

Micro Stack

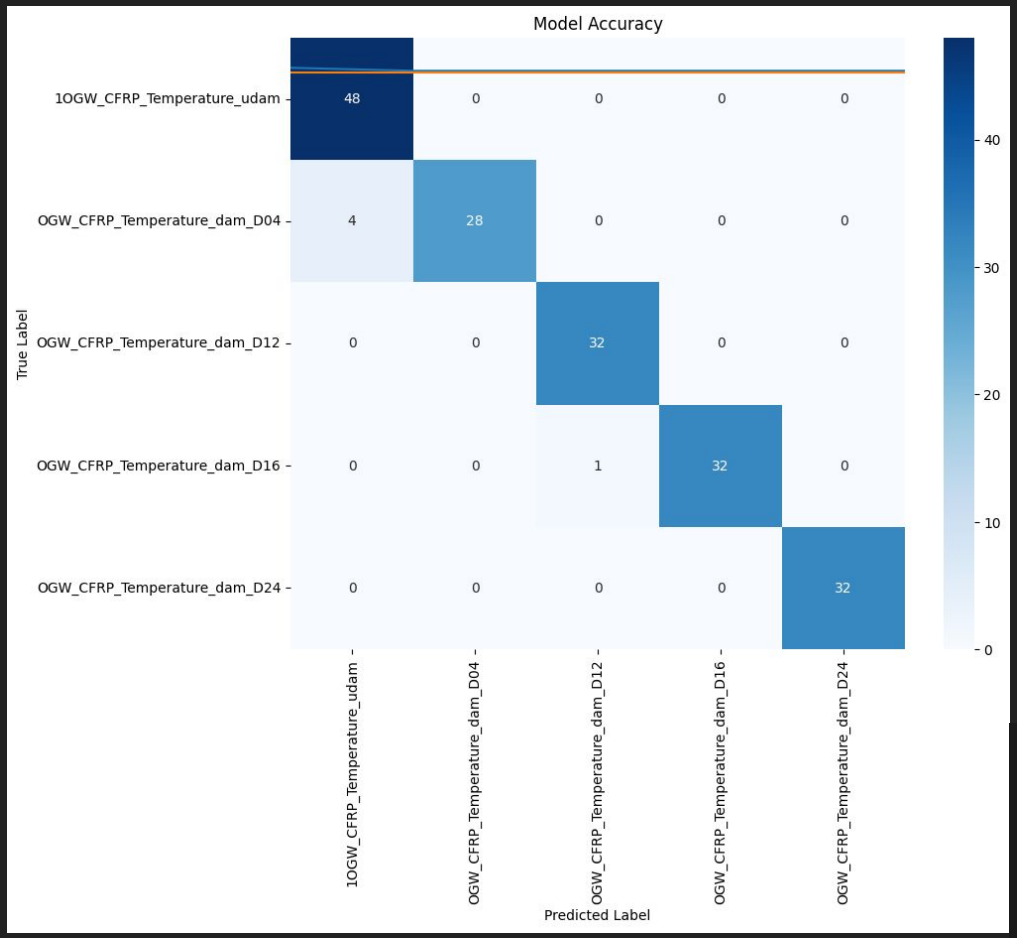


Figure 16: Micro Stack Model Confusion Matrix

Time to Train CPU
Only: ~8 minutes

Table 7: Micro Stack Statistics

	precision	recall	f1-score	support
0	0.92	1.00	0.96	48
1	1.00	0.88	0.93	32
2	0.97	1.00	0.98	32
3	1.00	0.97	0.98	33
4	1.00	1.00	1.00	32
accuracy			0.97	177
macro avg	0.98	0.97	0.97	177
weighted avg	0.97	0.97	0.97	177

Evaluation / Performance Analysis

- When comparing to [1], the accuracy of the models' stack and superstack are similar using similar data.
- Although they use Singular Value Decomposition to create their temporal robust damage index, which is converted from the 6 channels
- My Model instead uses 2D Convolutions to create a similar result
- Their time to train the model is less, although I only use the CPU, and I am unsure if they only use the CPU.

Conclusion

- The Resulting Models that I have constructed were effective at identifying damage and its location while accounting for temperature variations

Future Work / Recommendations

- Implementing the full convolutional stack to get "cross-correlation" for the Wave Speed
- Working to learn how to do Baseline Signal Stretch (BSS)
- Optimizing the number of parameters that reach 100% accuracy with smaller trainable parameters
- Working with PCA to get an unsupervised model working

References

[1]S. Zong, S. Wang, Z. Luo, X. Wu, H. Zhang, and Z. Ni, "Robust Damage Detection and Localization Under Complex Environmental Conditions Using Singular Value Decomposition-based Feature Extraction and One-dimensional Convolutional Neural Network," *Chinese Journal of Mechanical Engineering*, vol. 36, no. 1, May 2023, doi: https://doi.org/10.1186/s10033-023-00889-3.

[2]A. Mardanshahi, V. Nasir, S. Kazemirad, and M. M. Shokrieh, "Detection and classification of matrix cracking in laminated composites using guided wave propagation and artificial neural networks," *Composite Structures*, vol. 246, p. 112403, Aug. 2020, doi: https://doi.org/10.1016/j.compstruct.2020.112403.

[3]J. Moll *et al.*, "Open Guided Waves: online platform for ultrasonic guided wave measurements," *Structural Health Monitoring*, vol. 18, no. 5–6, pp. 1903–1914, Dec. 2018, doi: https://doi.org/10.1177/1475921718817169.

[4]IBM, "Support Vector Machine," *IBM*, Dec. 12, 2023. https://www.ibm.com/think/topics/support-vector-machine

[5]V. Giurgiutiu, "GUIDED WAVES," *Structural Health Monitoring*, pp. 185–238, 2008, doi: https://doi.org/10.1016/b978-012088760-6.50007-6.

[6]K. Diamanti and C. Soutis, "Structural health monitoring techniques for aircraft composite structures," *Progress in Aerospace Sciences*, vol. 46, no. 8, pp. 342–352, Nov. 2010, doi: https://doi.org/10.1016/j.paerosci.2010.05.001.

[7]IBM, "What is principal component analysis (PCA)?," *Ibm.com*, Dec. 08, 2023. https://www.ibm.com/think/topics/principal-component-analysis

[8]IBM, "Convolutional Neural Networks," *Ibm.com*, Oct. 06, 2021. https://www.ibm.com/think/topics/convolutional-neural-networks

References 2

[9] M. El Mountassir, S. Yaacoubi, and F. Dahmene, “Reducing false alarms in guided waves structural health monitoring of pipelines: Review synthesis and debate,” *International Journal of Pressure Vessels and Piping*, vol. 188, p. 104210, Dec. 2020, doi: https://doi.org/10.1016/j.ijpvp.2020.104210.

[10] GeeksforGeeks, “DBSCAN Clustering in ML Density based clustering,” *GeeksforGeeks*, May 06, 2019. https://www.geeksforgeeks.org/machine-learning/dbscan-clustering-in-ml-density-based-clustering/

[11] "scipy.signal.correlate — SciPy v1.7.0 Manual,” *docs.scipy.org*. https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.correlate.html

[12] A. Abbassi, N. Römgens, F. F. Tritschel, N. Penner, and R. Rolfes, “Evaluation of machine learning techniques for structural health monitoring using ultrasonic guided waves under varying temperature conditions,” *Structural Health Monitoring*, p. 147592172211075, Jun. 2022, doi: https://doi.org/10.1177/14759217221107566.

[13] S. SAWANT, S. BANERJEE, and A. SETHI, “DAMAGE IDENTIFICATION FOR GUIDED WAVE TESTING OF COMPOSITE STRUCTURES USING STATISTICAL FEATURES,” *Proceedings of the 14th International Workshop on Structural Health Monitoring*, Sep. 2023, doi: https://doi.org/10.12783/shm2023/36796.